

Group Topic: Big Data Analysis for Rainfall-Based Flood Risk Prediction in Albay using SYNOP Data	Lab No. 2
Git Repo/ Colab Link: https://github.com/PotatoKevz/About-bda	Date: 04/24/26

Objectives

- Conceptualize the data gathering method
- Look for online sites that can be requested for SYNOP data
- Decode the downloaded raw data reports in a CSV format
- Clean the decoded data reports
- Visualize the cleaned data reports

Tasks Completed

1. Looked for sites that can be requested by SYNOP Dataset, and the group arrived at OGIMET (*Organización Meteorológica Iberoamericana*), an online platform that provides access to historical weather observations, especially SYNOP reports from meteorological stations worldwide.

Professional information about meteorological conditions in the world

OGIMET

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Versión española

WEATHER MODEL FORECAST

GFS world weather maps

METEORGRAMS
Gramet aero
Gramet meteo

INDEXES
Query about indexes

UNDECODED REPORTS
Synop reports
Synop by countries
Latest synops
Metar/Taf reports
Latest metar
Latest taf
Upper air soundings
Soundings by countries

TEXT INFORMATION
Daily summaries
Country summaries

This is a Weather Information Service provided by www.ogimet.com. It uses freely available data from the net, mainly from [NOAA](#), and it uses Open Software to process it. The goal of this site is to provide data on current weather conditions in a fast and professional way. This is a free service in a narrow bandwidth server as you can see [here](#), so we ask you not to abuse it.

We are working to add new pages and services to this site. Our aim is to be useful

CONTENT INDEX

- [Queries about meteorological station indexes around the world](#). It uses both WMO and ICAO station indices, and allows efficient searches based on site, partial name string, country etc ...
- [Queries about meteorological reports](#):
 - [METAR and TAF queries](#). Web form page designed to query METAR, SPECI and TAF reports. You can select the stations, the order, time interval, format (text or html)...
 - [Latest METAR reports by country or territory](#). You can rapidly see latest METAR and TAF reports from a country. The reports are shown undecoded
 - [SYNOP report queries](#). Designed to answer queries about SYNOP and SYNOP SHIP reports. You can select the stations, the order, time interval and format (text or html)...
 - [Upper air soundings TEMP and PILOT reports](#). You can select reports by individual station or [by territory](#)
 - [Latest SYNOP reports by country or territory](#). Select the country to get the latest SYNOP reports received. The reports are shown undecoded
 - [Ogimet daily weather summaries](#). You can obtain a daily weather summary which is based on synop reports data, which has been available since July 2004. The result is a comprehensive daily summary data
 - [GSOD daily weather summaries](#). These summaries are based directly on [NOAA GSOD \(Global Summary Of the Day\)](#) reports. For some stations there is data from 1929. GSOD data became available in 1971 providing a world-wide coverage database
 - [Weather Ranking](#) A list of sites sorted by weather parameters. Selection is made by continent, country or territory
 - [CLIMAT monthly weather summaries](#). These summaries are based directly on CLIMAT reports (FM 71-KT), usually available during first 10 days of next month
- [METEOSAT / reports composition](#):
 - A [Meteosat-9 Infrared Image Satellite composition with metar](#) is available. The satellite Infrared µm image is combined with a precipitation estimation at [eumetsat](#). Then, if metar is available, a small circle with [metar's color](#) is plotted in the station position. When passing the mouse over the circle you can see the metar report. There is an image every 15 minutes
- [Weather map forecasts from GFS model](#). From basic grib data, selected parameters are plotted giving a world wide coverage of weather maps
- [Graphic information](#)
 - [Synop based information over geographic areas](#) It permits you to see decoded information over predetermined geographic zones. The information shown came from SYNOP and SYNOP SHIP reports
 - [Weather synop navigation](#). This page allows you to select a starting point and date. From it, you can go around the world and time get the decoded information from SYNOP and SYNOP SHIP reports over geographic areas. You can see station data when passing the mouse over the

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2. Scraped data from weather stations near Albay; Albay (Lezapi City), Camarines Norte (Daet), Camarines Sur (Naga), Catanduanes (Virac), Masbate, Romblon, and Quezon (Infanta and Tayabas) Stations:

	98427 Tayabas (Quezon) 75k.csv 	 jab2023-6181-30913	Apr 23	8.2 MB	
	98434 Infanta (Quezon) 75k.csv 	 jab2023-6181-30913	Apr 23	8.2 MB	
	98440 Camarines Norte 76k.csv 	 jab2023-6181-30913	Apr 23	8.5 MB	
	98442 Camarines Sur Station 11k.csv 	 jab2023-6181-30913	Apr 23	3.5 MB	
	98444 Albay Station 76k.csv 	 jab2023-6181-30913	Apr 23	8.4 MB	
	98446 Catanduanes 76k.csv 	 jab2023-6181-30913	Apr 23	8.2 MB	
	98536 Romblon 74k.csv 	 jab2023-6181-30913	Apr 23	7.9 MB	
	98543 Masbate 76k.csv 	 jab2023-6181-30913	Apr 23	8 MB	

Scraped Datasets from OGIMET

	A	B	C	D	E	F	G
1	WMOIND	YEAR	MONTH	DAY	HOUR	MIN	REPORT
2	98444	2000	1	1	0	0	AAXX 01001 98444
3	98444	2000	1	1	3	0	AAXX 01031 98444
4	98444	2000	1	1	6	0	AAXX 01061 98444
5	98444	2000	1	1	9	0	AAXX 01091 98444
6	98444	2000	1	1	12	0	AAXX 01121 98444
7	98444	2000	1	1	15	0	AAXX 01151 98444
8	98444	2000	1	1	18	0	AAXX 01181 98444
9	98444	2000	1	2	0	0	AAXX 02001 98444
10	98444	2000	1	2	3	0	AAXX 02031 98444
11	98444	2000	1	2	6	0	AAXX 02061 98444
12	98444	2000	1	2	9	0	AAXX 02091 98444
13	98444	2000	1	2	12	0	AAXX 02121 98444
14	98444	2000	1	2	15	0	AAXX 02151 98444
15	98444	2000	1	2	18	0	AAXX 02181 98444
16	98444	2000	1	2	21	0	AAXX 02211 98444
17	98444	2000	1	3	0	0	AAXX 03001 98444
18	98444	2000	1	3	3	0	AAXX 03031 98444
19	98444	2000	1	3	6	0	AAXX 03061 98444
20	98444	2000	1	3	9	0	AAXX 03091 98444
21	98444	2000	1	3	12	0	AAXX 03121 98444
22	98444	2000	1	4	0	0	AAXX 04001 98444
23	98444	2000	1	4	3	0	AAXX 04031 98444
24	98444	2000	1	4	6	0	AAXX 04061 98444
25	98444	2000	1	4	9	0	AAXX 04091 98444
26	98444	2000	1	4	12	0	AAXX 04121 98444
27	98444	2000	1	4	15	0	AAXX 04151 98444

CSV Content of Albay Station (Raw SYNOT Data Report)

```
C:\Users> romeo > Desktop > Big Data Analysis > Programs > flood-prediction > ogimet_synop_scraper.py > _chunks
56 BASE_URL = "http://www.ogimet.com/cgi-bin/getsynop"
57 COLUMNS = ["WMOIND", "YEAR", "MONTH", "DAY", "HOUR", "MIN", "REPORT"]
58
59 logging.basicConfig(
60     level=logging.INFO,
61     format="%(asctime)s [%(levelname)s] %(message)s",
62     datefmt="%Y-%m-%d %H:%M:%S",
63 )
64 log = logging.getLogger(__name__)
65
66 # Mimic a real browser - OGIMET rejects bot-like user agents
67 HTTP_HEADERS = {
68     "User-Agent": (
69         "Mozilla/5.0 (Windows NT 10.0; Win64; x64) "
70         "AppleWebKit/537.36 (KHTML, like Gecko) "
71         "Chrome/124.0.0 Safari/537.36"
72     ),
73     "Accept": "text/plain, text/html, */*",
74     "Accept-Language": "en-US,en;q=0.9",
75     "Connection": "keep-alive",
76 }
77
78
79 # --- helpers ---
80
81 def _fmt(dt: datetime) -> str:
82     return dt.strftime("%Y%m%d%H%M")
83
84
85 def _chunks(start_year: int, end_year: int, days: int) -> Iterator[Tuple[datetime, datetime]]:
86     """Yield (begin, end) datetime pairs in `days`-sized steps."""
87     now = datetime.now(timezone.utc).replace(tzinfo=None)
```

Code Snippet of Data Scraper

3. Cleaned the decoded SYNOP datasets:

 CleanedAlbay.csv 	 jab2023-6181-30913	6:37 PM	5.4 MB	
 CleanedCamNorte.csv 	 jab2023-6181-30913	6:42 PM	5.4 MB	
 CleanedCatanduanes.csv 	 jab2023-6181-30913	6:42 PM	5.3 MB	
 CleanedInfanta.csv 	 jab2023-6181-30913	6:43 PM	5.3 MB	
 CleanedMasbate.csv 	 jab2023-6181-30913	6:43 PM	5.4 MB	
 CleanedRomblon.csv 	 jab2023-6181-30913	6:43 PM	5.2 MB	
 CleanedTayabas.csv 	 jab2023-6181-30913	6:44 PM	5.3 MB	

Cleaned Datasets

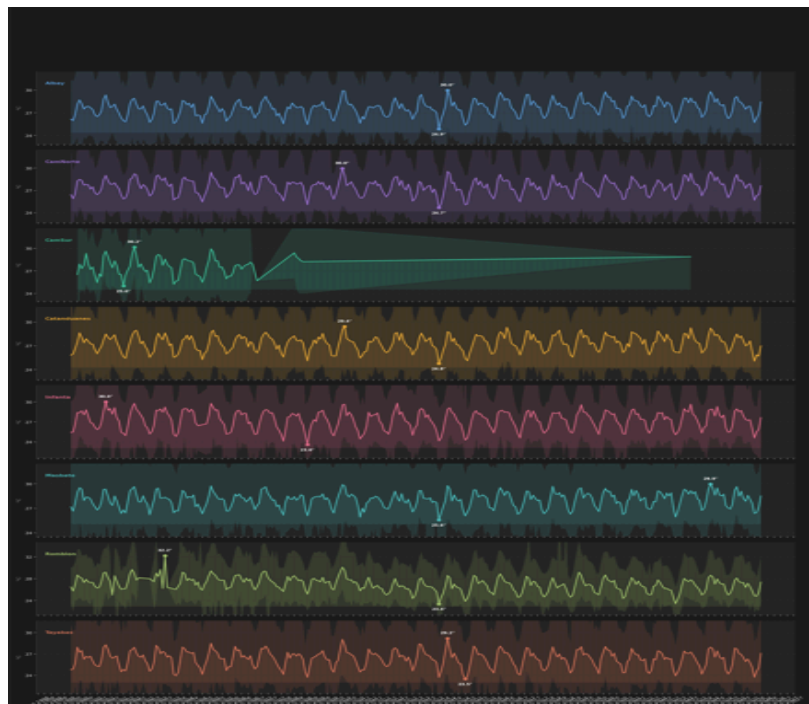
```

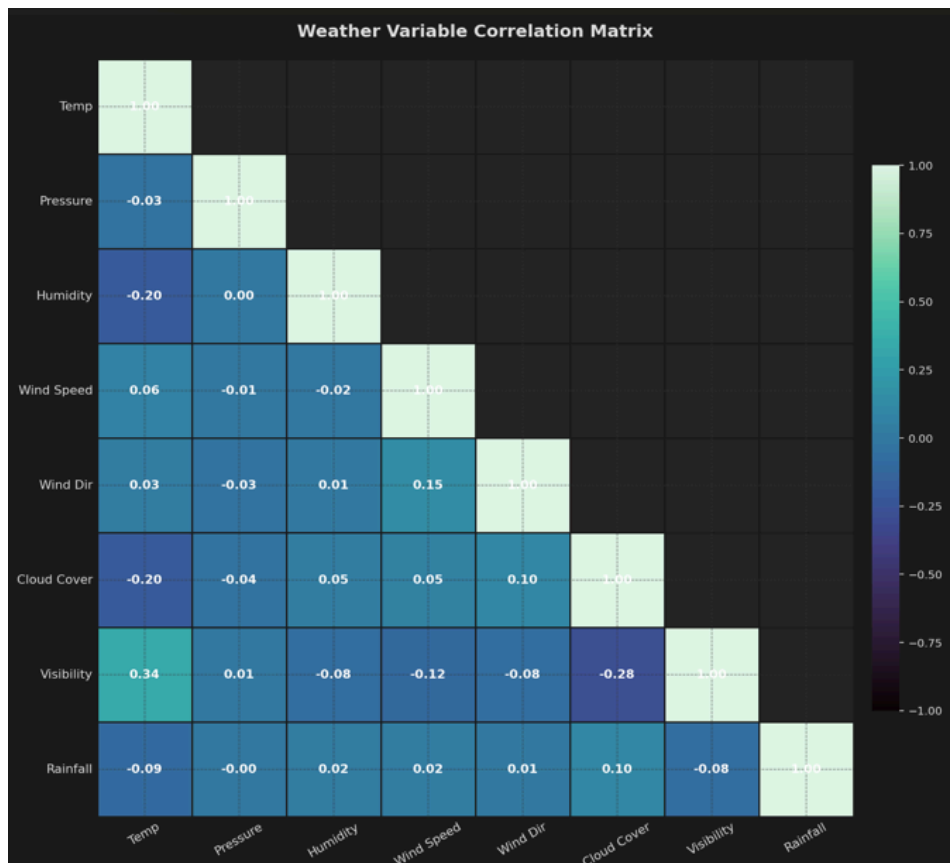
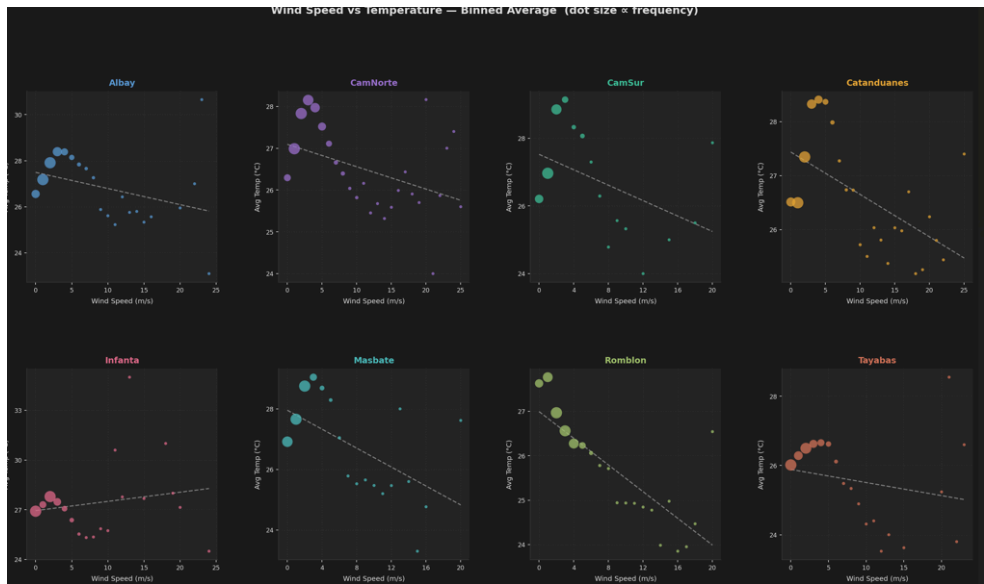
C: > Users > romeo > Desktop > Big Data Analysis > Programs > flood-prediction > synop_decoder.py > ...
62
63 def decode_iriix(group: str) -> dict:
64     """Section 0 indicator group iRiXhVW (5 chars).
65
66     iR (precipitation indicator):
67         0/1/2 group 6 present somewhere → no sentinel (use decoded value)
68         3 precip amount = 0, no group 6 → rain_sentinel = 0.0 (Case B)
69         4 station does not measure precip → rain_sentinel = None (Case A)
70     """
71     out = {}
72     if len(group) != 5:
73         return out
74     ir = group[0]
75     if ir == "3":
76         out["rain_sentinel"] = 0.0 # "no rain reported" → 0.0 mm
77     elif ir == "4":
78         out["rain_sentinel"] = None # truly missing (station doesn't measure)
79     # iX - station type / present-weather indicator (ignored)
80     # h - height of cloud base (code table 1600)
81     h_code = group[2]
82     h_table = {"0": 0, "1": 50, "2": 100, "3": 200, "4": 300,
83               "5": 600, "6": 1000, "7": 1500, "8": 2000, "9": 2500, "/": None}
84     out["cloud_base_m"] = h_table.get(h_code)
85     # VV - visibility (code table 4377)
86     vv = group[3:5]
87     out["visibility_m"] = _decode_vv(vv)
88     return out
89
90
91 def _decode_vv(vv: str) -> float | None:
92     """Convert VV code (00-99) to metres (WMO Code Table 4377)."""

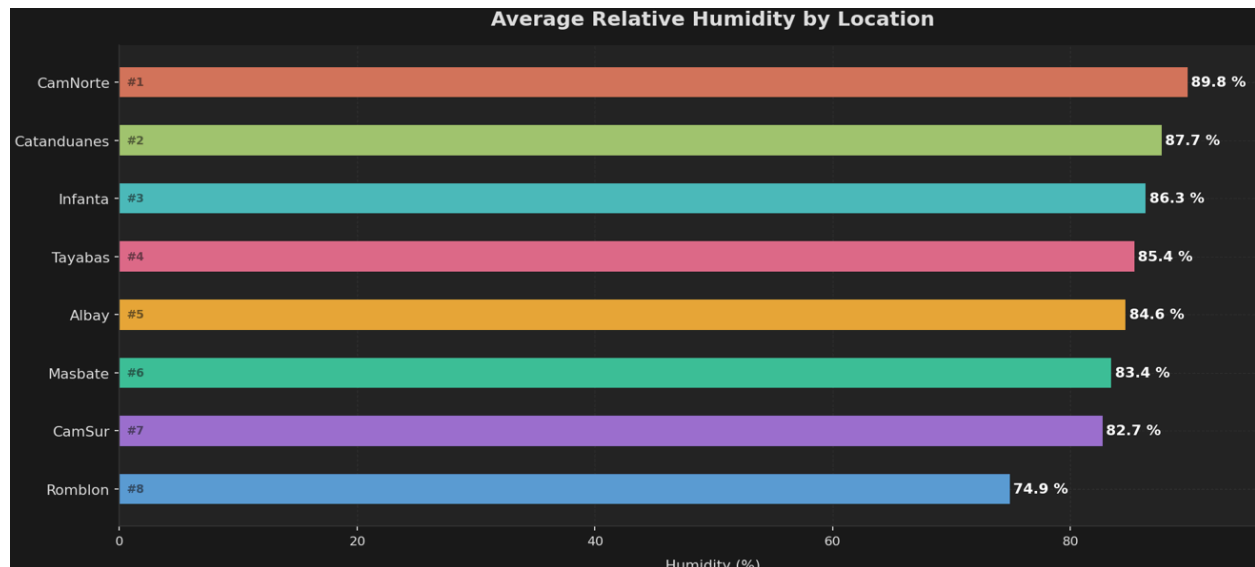
```

Code Snippet

4. Visualization of the datasets:







Results and Analysis

Summary of Findings

- **Thermal Stability:** All eight stations maintain a consistent tropical range between 26.3°C (**TAYABAS**) and 28.0°C (**MASBATE**), with a clear seasonal peak in May - June (~28.7°C).
- **Regional Anomalies:** January 2014 was identified as a significant regional cold event, showing record lows across five different stations simultaneously.
- **Humidity & Geography:** Humidity levels are high across the mainland (82% - 90%) but **ROMBLON** acts as a significant outlier **74.9%**, likely due to its island geography and rain-shadow effects.
- **Wind Patterns:** Mean daily wind speeds peak during the Northeast Monsoon (Amihan) in December and January. The data accurately reflected extreme weather, such as the **60.0m/s** gust recorded in Camarines Sur in September 2000.

Challenges and Solutions

Challenge: Difficulty of coming up with ways to visualize the data from the cleaned csv field (what the logic will be).

Solution: Making use of AI and having it give out examples as to how this can be done.

Challenge: Clarity of the visualizers are not being outputted the same way that I had in mind.

Solution: Adjusting the color palette and the theme that the graphs output, and also the orientation of the data being outputted.

Conclusion

This week's tasks successfully moved the project from the planning stage to active data management. By establishing an automated pipeline to scrape and decode SYNOP data, the group has secured the necessary high-fidelity information required for rainfall-based flood prediction. The objectives for data gathering, decoding, and initial visualization were fully met